

DEEPAK PRAJAPATI

AI/ML Engineer | Generative AI & Agentic AI Systems

deepakprajapati27131@gmail.com | +91 8922828060 | [LinkedIn](#) | GitHub: [Deepak27131](#)

PROFESSIONAL SUMMARY

AI/ML Engineer specializing in Agentic AI systems and RAG pipelines, with hands-on experience across Machine Learning, Deep Learning, and NLP. Built and deployed 3+ end-to-end GenAI applications — including multi-agent research and travel-planning assistants with human-in-the-loop workflows — from prototype to live, publicly accessible demos on AWS. Seeking entry-level opportunities in AI/ML Engineering, Agentic AI, or AI Consulting to design and ship scalable intelligent systems.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy

Machine Learning: Regression, Classification, Clustering, Model Evaluation

Deep Learning: ANN, CNN, LSTM, TensorFlow, PyTorch

Generative AI & Agentic AI: LLMs, Prompt Engineering, RAG, LangChain, LangGraph, Multi-Agent Systems, Model Context Protocol (MCP), FAISS, ChromaDB, OpenAI APIs, Gemini APIs, Mistral AI, Hugging Face

MLOps & Deployment: FastAPI, Docker, AWS (EC2), Nginx, systemd, Render, Git

Visualization & BI: Power BI, Excel, Matplotlib

Automation: n8n

PROFESSIONAL EXPERIENCE

AI Engineer Trainee | Regex Software And Services

March 2026 – Present

- Performed data cleaning and EDA on structured datasets with 10,000+ records, identifying key trends that directly informed AI model development decisions.
- Designed and built automated ETL and preprocessing pipelines in Python and SQL, streamlining data ingestion and reducing manual processing effort.
- Designed interactive Power BI dashboards and KPI reports to support performance monitoring and data-driven decision-making.

AI Intern (Virtual Internship) | IBM

January 2026 – February 2026

- Built a RAG-based AI chatbot using LLMs, LangChain, and FAISS with prompt engineering for accurate, document-grounded answers; developed an NLP-based Text Emotion Detection system to classify sentiment from raw text.

KEY PROJECTS

PlanMyTrip AI – Multi-Agent Travel Planning System (Python, LangGraph, LLMs, MCP, Streamlit, AWS EC2)

Jun – Aug 2026

- Engineered a supervisor-orchestrated, 7-agent system (Supervisor, Flight, Hotel, Weather, Budget, Itinerary, Final Response) using LangGraph, with an input guardrail that validates requests before any downstream agent or tool call runs.
- Implemented a human-in-the-loop approval workflow using LangGraph's interrupt/resume pattern, letting users review and revise AI-generated itineraries before finalization.
- Integrated 3 external data sources via Model Context Protocol (MCP) servers — Tavily Search, AviationStack, and OpenWeather — for real-time flight, hotel, and weather data.
- Built a Streamlit chat interface with PostgreSQL-backed authentication and persistent multi-turn state via LangGraph's Postgres checkpoint; deployed on AWS EC2 with systemd and an Nginx reverse proxy. Live demo: 13.63.62.113:8501

ResearchMind – Multi-Agent AI Research Assistant (Python, LLMs, LangChain, Streamlit, AWS EC2)

Jan – Mar 2026

- Designed a collaborative 5-agent architecture (Research, Analysis, Summarization, Response, and Orchestrator agents) that automates the end-to-end research workflow — from information gathering to structured report generation — reducing manual research effort.
- Built the Orchestrator Agent to coordinate task hand-offs and data flow between agents, enabling context-aware retrieval and AI-powered summarization in a single streamlined pipeline via an interactive Streamlit interface.
- Deployed on AWS EC2 in a production-style Linux environment with a live, publicly accessible demo: 13.62.58.130:8501
- GitHub: [Deepak27131/ResearchMind-MultiAgent](#)

Insurance Premium Prediction System (Python, Scikit-Learn, FastAPI, Streamlit, Docker, AWS EC2)

Apr – May 2026

- Built an end-to-end ML deployment pipeline: trained a Scikit-Learn model to predict insurance premium categories, served it via a FastAPI REST API (with Swagger docs), and connected a Streamlit frontend.
- Containerized both services with Docker (pushed to Docker Hub) and deployed on a public AWS EC2 instance, demonstrating a production-style MLOps workflow from model to public API.
- Live: UI – 13.51.79.228:8501

CERTIFICATIONS & ACHIEVEMENTS

- AI Engineering Certificate – IBM | Data with Generative AI – Udemy
- Data Analytics Job Simulation – Deloitte (April 2025)
- HackerRank: 5-Star Badge in SQL & Python | Solved 100+ DSA problems on LeetCode

EDUCATION

B.Tech in Computer Science and Engineering

July 2022 – June 2026

Vivekananda Global University, Jaipur

CGPA: 8.21 | Degree conferral: January 2027 | Coursework: Machine Learning, Data Mining, Statistics, DSA